

COMPREHENSIVE FINDINGS REPORT — PHASES I THROUGH
XIV

UBP Harmonic Study: What We Found

A synthesis of fourteen phases of investigation into the Unified Binary Physics system and its relationship to musical harmony, prime number topology, and the structural limits of coding-theoretic approaches to consonance.

Scope Music Theory / Coding Theory / Number Theory

Corpus 14 Phases, 37 Core Tests, 2000+ Integers

Date July 2026

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1. Executive Summary

This report synthesises the findings of fourteen phases of investigation into the Unified Binary Physics (UBP) system and its relationship to musical harmony and prime number structure. The study began as a computational musicology experiment, asking whether the UBP's error-correcting codes (Golay [24,12,8], Leech lattice 24D, Barnes-Wall 256D) could naturally encode the perceptual hierarchy of musical consonance. It evolved into a broader investigation of whether topological and coding-theoretic structures carry information about prime numbers, factorization, and the deep interplay between number theory and the physics of harmony.

Across 14 phases, the study produced several rigorous results, a number of definitive negative findings, and a small set of genuine discoveries that survive honest stress-testing. The most important single finding is that the consonance signal lives not in the UBP's coding theory but in number theory itself: pure JI exponent vectors achieve $r = +0.9613$ for interval consonance, and the Mersenne/Fermat mod-144 duality provides a structural explanation for why 12-tone equal temperament decomposes into a Mersenne number (the fifth, 7 semitones) plus a Fermat prime (the fourth, 5 semitones). This is a mathematical identity, not an artifact of the UBP system, and it is invariant under modulus scaling.

For the UBP system specifically, the study established that the Golay code has a fundamental "three-distance ceiling" (inter-codeword distances of 8, 12, and 16) that makes chord-level differentiation mathematically impossible within the coding layer. No amount of dimensional expansion through the Leech lattice or Barnes-Wall lattice overcomes this constraint. However, a separate "Prime-Layer Harmonic Module" using Jaccard distance on Mersenne/Fermat residue proximity sets achieves $r = +0.8244$ for chord consonance, operating independently of the error-correction layer. On the prime number front, the study found that rotation sign changes in Lucas-Lehmer iteration correlate at $r = +0.797$ with factor count, but attempts to estimate actual factor magnitudes from spectral features failed on general composites, succeeding only on Mersenne-specific structures.

2. Confirmed Strengths of the UBP System

2.1 Interval Consonance via Circle-of-Fifths Gray Encoding

The strongest positive result for the UBP coding layer is the interval-level consonance correlation. When pitch classes are encoded using Gray codes derived from the Circle of Fifths ordering and then passed through the Golay [24,12,8] encoder, the Hamming distances between codeword pairs achieve a Pearson correlation of $r = +0.8674$ with empirical consonance ratings ($R\text{-squared} = 75.2\%$). This works because the CoF ordering creates Gray-code adjacency relationships that map naturally onto the Golay code's distance structure, producing a three-step causal chain: Circle-of-Fifths ordering produces Gray adjacency, which maps to Golay octad distances, which correlate with consonance. The encoding is not unique (ranking only in the top 3.3% of 50,000 random permutations), but it is genuine and reproducible.

The CoF ordering itself, the use of Gray codes (without which the correlation drops from $r = 0.87$ to $r = 0.18$), and the selection of the Golay code as the coding layer are all essential ingredients that contribute to this result.

2.2 Mersenne/Fermat Mod-144 Duality (Mathematical Identity)

A deep number-theoretic discovery emerged from Phase VIII: Mersenne numbers ($2^p - 1$ for prime p) modulo 144 always reduce to residues in the set $\{31, 127\}$, while Fermat numbers ($2^{2^k} + 1$ for $k \geq 2$) always reduce to $\{17, 113\}$. These four residues satisfy the XOR identity $31 \text{ XOR } 127 = 17 \text{ XOR } 113 = 96$, which equals $2^5 + 2^6 =$ two-thirds of 144. More importantly, every Mersenne residue is congruent to 7 modulo 12 (the perfect fifth in semitone notation), and every Fermat residue is congruent to 5 modulo 12 (the perfect fourth). This means the fifth/fourth structure is baked into the number theory itself, not specific to the choice of modulus. Phase XI proved this is an invariance theorem: the correlation remains exactly $r = -0.3770$ across all 11 tested moduli from 144 to 2,985,984. The practical consequence is that $12\text{-TET} = \text{fifth}(7) + \text{fourth}(5) = \text{Mersenne} + \text{Fermat}$, a structural decomposition of the octave that connects tuning theory directly to the theory of special prime-generating functions.

2.3 The Prime-Layer Harmonic Module

The 4D Prime Residue Fingerprint, defined as $[d(17), d(31), d(113), d(127)]$ measuring modular distances to the four anchor residues in mod-144 space, provides the best chord consonance signal found in the entire study. Using Jaccard distance on threshold-10 proximity sets derived from this fingerprint, the system achieves $r = +0.8244$ for chord consonance, the first positive chord correlation above 0.5 in any phase. The mechanism is a natural 7/5 partition created by the residues in mod-144 space: residue 17 sits at position 17, creating a boundary between F-sharp and G that corresponds exactly to the tritone-fifth boundary in musical pitch space. Consonant chords (major and minor triads) consistently span this boundary, while dissonant clusters tend to stay on one side. This mapping is perfectly transposition-invariant: all 12 major triads produce identical Jaccard values (0.3333), as do all 12 minor triads. The prime layer operates entirely independently of the Golay/Leech error-correction layer, suggesting a clean architectural separation: the coding layer handles error correction and stability, while the prime layer handles harmonic structure.

2.4 Rotation Sign Changes and Factor Count

Phase XII discovered that the number of sign changes in the 4D rotation direction signal during Lucas-Lehmer iteration correlates at $r = +0.797$ with the factor count of the Mersenne number being tested. This is the strongest topological signal found for primality-related properties. The rotation signal tracks how the 4D residue fingerprint "turns" in its ambient space as the LL recurrence $s = s^2 - 2 \pmod{M_p}$ progresses through its iterations. Numbers with more distinct prime factors produce more frequent direction reversals, creating a measurable topological signature of factorization complexity. While this signal is real, it is important to note that it was tested primarily on Mersenne composites (where the LL recurrence has special mathematical structure), and generalization to arbitrary composites requires further investigation.

2.5 Dimensional Hierarchy Preservation

The interval consonance correlation of $r = +0.8674$ is perfectly preserved across all dimensional expansions tested: Golay(24D), Barnes-Wall(256D), Barnes-Wall(512D), and Barnes-Wall(1024D) all produce exactly the same correlation. The absolute distance values scale with dimension (from range [16.0, 19.6] at 256D to [32.0, 39.2] at 1024D), but the rank ordering of interval distances remains invariant. This means the UBP's dimensional expansion mechanism, while unable to enrich the harmonic signal beyond the Golay ceiling, also does not degrade it. The signal is dimensionally stable, which is a positive architectural property for any system that needs to operate across multiple representation scales without losing information.

3. Definitive Negative Results

3.1 The Three-Distance Ceiling (Fundamental Limitation)

The most consequential negative finding is that the Golay [24,12,8] code has exactly three possible inter-codeword Hamming distances: 8, 12, and 16. In practice, only two of these (8 and 12) appear for the CoF Gray encoding of 12 pitch classes. This means the coding layer can only distinguish two levels of interval consonance, while music theory requires at least six distinct ranks (unison, octave, fifth/fourth, thirds/sixths, seconds/sevenths, tritone). Phase III confirmed this with a complete failure: all triads map to weight-8 octads regardless of their consonance, producing $r = 0.00$ for every chord metric tested. No non-linear aggregation method (AND gate, majority vote, NRCI arithmetic/geometric/harmonic mean, OR gate, Leech cloud statistics) overcame this ceiling. The 3-distance constraint is a property of the Golay code itself, not of the encoding scheme, and it is mathematically absolute. Any UBP application that requires fine-grained chord differentiation through the coding layer alone will fail.

3.2 Leech Lattice Expansion Destroys the Signal

While the Barnes-Wall expansion preserves the interval correlation (because it inherits the Golay structure), the Leech lattice 24D expansion destroys it. The Euclidean distance between Leech cloud centroids for different intervals produces only $r = 0.13$, down from $r = 0.87$ at the Golay level. The reason is that the 128-point symmetric expansion used to embed Golay codewords into Leech lattice vectors creates point clouds whose centroids all converge to zero (the mean of any symmetric set of Leech points is the origin). Individual cloud distributions are nearly identical across all intervals (mean approximately 7.93, mode 8.0 for every interval), washing out the binary distance signal entirely. The Leech lattice's extraordinary error-correction properties come at the cost of collapsing the very structure that carries the harmonic information. However, Phase IX did find that the active dimension sets (which specific coordinates are non-zero) preserve the signal at $r = 0.8429$ for intervals, suggesting that the information survives in the pattern of which dimensions are used, not in the Euclidean distances between points.

3.3 Range-Dependent Inversion of Metrics

Phase XIV revealed a critical instability in the UBP primality metrics: their correlation with primality changes sign depending on the numeric range being examined. NRCI and Topological Shear Gamma are strong predictors ($|r| > 0.3$) in the range 1000-3500 but collapse to near-zero above 4000. The rotation sign-change metric inverts four separate times across the 1000-10000 range, at bands 1000-2000, 4000-5000, 6000-7000, and 7000-8000. Syndrome weight (snapped) shows zero correlation in all bands. The only metric that remains stable across all ranges is anchor distance, which maintains r between $+0.312$ and $+0.382$ with a mean of $+0.339$. This means any primality test or factorization heuristic built on UBP metrics must either be range-calibrated or restricted to anchor distance as the sole reliable indicator. The range-dependent inversion is a serious obstacle to building general-purpose algorithms from these signals.

3.4 Factor Size Estimation Failure

Phase XIII initially found $r = +0.7963$ between $\log(\text{max factor})$ and the spectral centroid of the rotation signal on five Mersenne composites, suggesting that large factors create a "low-frequency hum" in the rotation trajectory. However, Phase XIV tested this on 134 general composites (semiprimes, three-factor numbers, prime powers, and general composites) and found that the maximum correlation between any spectral feature and $\log(\text{max factor})$ was only $r = 0.144$. After controlling for $\log(n)$ via partial correlation, all factor-spectral correlations dropped below $|r| = 0.29$. A ridge regression model trained on the general composites achieved test $r = +0.202$ for $\log(\text{max factor})$ and $r = +0.295$ for $\log(\text{min factor})$, both far too weak for practical estimation. On a blind test using Mersenne composites, the model regressed completely to the mean, predicting all max factors at approximately 160. The conclusion is that the spectral features primarily encode the magnitude of the number ($\log n$), not its factorization structure. The Phase XIII finding was real but domain-specific to Mersenne numbers, where the $s^2 - 2$ recurrence has special algebraic properties that do not exist for general moduli.

3.5 The Tenacity Law Hypothesis

The UBP Prime document hypothesised that "Sovereign Primes" exist at global minima of the Lock Pressure landscape, with composites at local maxima ("zero-pressure ghosts"). Phase XIII mapped the full Lock Pressure landscape across 9000 numbers in the range [1000, 10000) and found the opposite pattern: primes have higher mean Lock Pressure (0.092 vs 0.039 for composites) but lower pressure percentile (0.267 vs 0.121). Fully 80.9% of composites are zero-pressure ghosts ($P = 0$), compared to only 56.6% of primes. The Topological Shear Gamma threshold of 0.008 passes 99.9% of primes but also 100% of composites, providing no separation. The stabilisation energy ratio between primes and composites is only 1.04x, effectively no difference. In this range at least, the simple Tenacity Law hypothesis does not hold. Whether it manifests at larger scales remains an open question, but the current evidence does not support the existence of Sovereign Primes as defined.

3.6 The Low-Friction Attractor and Friction Ratio

Phase XI found that composites generate 5.6x more Cayley-Menger hypervolume deformation and 3.4x more geometric rotation during LL iteration, suggesting that composites experience more "topological friction." Phase XII tested this on the broader landscape of 2000 numbers and discovered a critical inversion: the finding was Mersenne-specific. In the general landscape, primes have higher friction and lower NRCI than composites, the opposite of the Mersenne-specific pattern. The friction ratio (friction / NRCI) as a universal primality metric achieved an AUC of only 0.309, worse than random guessing. The NRCI scaling law showed decay at -0.000110 per bit, confirming that the coding-layer primality signal weakens at higher bit widths. The Golay-Leech normalisation filter (Directive 3 of Phase XIII) produced a slope of -0.000112, virtually identical to the raw slope, confirming that normalisation does not prevent signal decay.

4. The Golay Three-Distance Ceiling: A Structural Explanation

The three-distance ceiling is not merely an empirical observation but a structural consequence of the Golay code's mathematical properties. The extended binary Golay code [24, 12, 8] is a perfect code: it has minimum distance 8, and its weight enumerator polynomial dictates that the possible weights of codewords are 0, 8, 12, 16, and 24. When 12 pitch classes are encoded as distinct codewords, the inter-codeword Hamming distances can only take values from the set {0, 8, 12, 16, 24}. In practice, the CoF Gray encoding maps the 12 pitches to codewords that differ by distances of 8 and 12 only (the distance-16 and distance-24 pairs do not occur in this particular encoding). This creates a binary classification: intervals are either "close" (distance 8) or "far" (distance 12) in Golay space. Since there are six consonance categories but only two distance buckets, the Golay code acts as a lossy compressor that collapses the harmonic hierarchy into a 1-bit signal.

The reason the interval correlation is still strong ($r = 0.87$) is that the two distance buckets happen to align roughly with a consonant/dissonant split: perfect fifths, fourths, and unisons tend to fall in one bucket while minor seconds, major sevenths, and tritones tend to fall in the other. But this alignment is coarse and does not extend to chord-level analysis, where the permutations of interval distances within a chord all map to the same set of {8, 12} values. The implication for the UBP system is clear: the error-correction layer was designed for reliability, not for harmonic differentiation. Any application requiring fine-grained harmonic analysis must look beyond the coding layer to the prime number structures that underlie the tuning system itself.

Metric	Correlation	Domain	Status
JI Exponent Intervals	$r = +0.9613$	Number theory	Strongest interval signal
Jaccard prox_10 Chords	$r = +0.8244$	Prime residue	Strongest chord signal
CoF Gray Golay Intervals	$r = +0.8674$	Coding layer	Strong interval, no chord

4D Fingerprint Chords	$r = -0.8790$	Prime residue	Euclidean, inverted
Rotation Sign Changes	$r = +0.797$	Topological	Mersenne-specific
Spectral Centroid (Mersenne)	$r = +0.7963$	Frequency domain	Mersenne-only
Anchor Distance	$r = +0.339$	Coding layer	Band-invariant
Friction Ratio AUC	0.309	Manifold	Below random
Factor Size Estimation	$r = +0.202$	Regression	No predictive power

Table 1. Summary of key correlations across all 14 phases

5. Prime Number Structure: What Survives Stress-Testing

5.1 The Mersenne/Fermat Invariance Theorem

The single most robust finding across the entire study is the invariance theorem for Mersenne and Fermat residues. For all primes $p \geq 5$, the Mersenne number $2^p - 1$ modulo 144 falls into the set $\{31, 127\}$, and both of these residues are congruent to 7 modulo 12. For all $k \geq 2$, the Fermat number $2^{2^k} + 1$ modulo 144 falls into $\{17, 113\}$, and both are congruent to 5 modulo 12. This is not a coincidence of the modulus 144 but a consequence of the fact that $2^p \bmod 12$ cycles through $\{4, 8\}$ for $p \geq 2$, and therefore $2^p - 1 \bmod 12$ is always 3 or 7. For $p \geq 5$, the modulus 144 constrains the result further to $\{31, 127\}$ (both congruent to 7). Similarly, $2^{2^k} \bmod 12 = 4$ for all $k \geq 2$, so $2^{2^k} + 1 \bmod 12 = 5$ always. This means the fifth (7 semitones) and fourth (5 semitones) are built into the binary expansion properties of Mersenne and Fermat numbers, and this structure is invariant under any choice of modulus that is a multiple of 12. The modulus $144 = 12^2$ is the minimal modulus that avoids wrap-around artifacts (since the largest residue, 127, plus one octave span of 12, gives $139 < 144$).

5.2 The XOR Bridge: 96 = Two-Thirds of 144

The XOR identity $31 \text{ XOR } 127 = 17 \text{ XOR } 113 = 96$ is a binary-level connection between the Mersenne and Fermat families. In binary, 96 = 01100000, which occupies bits 5 and 6 (the two middle-high bits of an 8-bit representation). This is exactly two-thirds of 144, placing it at a natural "bridge point" in the mod-144 space. The Lucas-Lehmer sequence always starts at $s = 4$, and the 4D residue fingerprint of 4 has a total distance sum of exactly 96, meaning the LL iteration begins at the Mersenne/Fermat bridge point regardless of which Mersenne number is being tested. This is a structural constant of the LL test, not a property of individual primes or composites. The XOR bridge also explains why all four prime residues produce identical UBP metrics (Hamming weight 8, NRCI 0.762346, Tax 3.1174): despite

having different residue values, they are related by XOR operations that preserve the coding-theoretic properties measured by the UBP system.

5.3 The 4D Fingerprint and Prime-Layer Module Specification

The 4D Prime Residue Fingerprint, defined as the vector $[d(17), d(31), d(113), d(127)]$ where $d(r, n)$ is the modular distance from n to residue r in mod-144 space, provides a complete specification for a Prime-Layer Harmonic Module that operates independently of the Golay/Leech coding layer. The module has three components: (1) Pitch encoding uses the 4D fingerprint vector directly. (2) Interval metrics use Jaccard distance on binary feature sets derived from the fingerprint, achieving $r = 0.76$ with 18 unique distance values (compared to only 2 for the Golay layer). (3) Chord metrics use Jaccard distance on residue proximity sets with threshold 10, achieving $r = 0.82$ with perfect transposition invariance. The module requires no error-correction machinery, operates in four dimensions rather than 24 or more, and its computational cost is negligible compared to the full Golay-Leech pipeline. The question of whether this module should be formally integrated into the UBP system architecture is perhaps the single most consequential architectural decision emerging from this study.

6. Failed Hypotheses and the Entropy Horizon

6.1 Music Cannot Predict Primes

The Entropy Horizon, as defined in Phase XI, is the fundamental limit at which the information content of musical structures (which live in a 12-dimensional space with severe symmetries) becomes insufficient to distinguish the much more complex structure of prime numbers (which live in an effectively infinite-dimensional space). The 4D residue fingerprint cannot sieve primes from composites ($r = -0.0094$), UBP noise does not cluster in dissonant spectral space, and the LL NRCI trajectories show only subtle differences that do not produce clean separation. The pigeonhole principle provides the theoretical foundation: with only 144^4 possible 4D fingerprint states but infinitely many primes and composites, any primality signal in this space must eventually saturate. The "Entropy Horizon" is the point at which this saturation makes further discrimination impossible. In practice, the UBP coding layer does carry a genuine primality signal (primes have higher NRCI of 0.921 vs composites at 0.735, and higher syndrome weight of 3.43 vs 6.07), but this signal lives in the error-correction machinery, not in the harmonic structure.

6.2 NRCI Scaling Law Decay

The NRCI primality signal decays at a rate of -0.000110 per additional bit of encoding width. At 12 bits (the standard UBP encoding), the signal is modest; at 64 bits, it is nearly gone. The Golay-Leech normalisation does not prevent this decay (normalised slope = -0.000112). However, Phase XIII discovered that the rank-order correlation ($r = -0.407$ between the binary prime/composite label and NRCI) is perfectly invariant across all tested bit widths from 12 to 64. This means the relative ordering of

primes versus composites is preserved even as the absolute signal strength converges toward 1.0. The scaling law is not in the absolute signal strength but in the relative order, which constitutes a genuine (if weak) structural property of the UBP encoding. Whether this rank-order invariance extends to much larger bit widths (hundreds or thousands of bits) remains untested and is an important open question.

6.3 Summary of What Did Not Work

Hypothesis	Phase	Result	Reason
Chord XOR synthesis	III	$r = 0.000$	All triads map to weight-8 octads
Leech Euclidean intervals	VI	$r = 0.13$	128-point symmetric expansion washes signal
Non-linear chord aggregation	IX	All $ r < 0.25$	Cannot escape 3-distance ceiling
Prime sieving via 4D fingerprint	XI	$r = -0.009$	Pigeonhole principle saturation
Noise spectral clustering	XI	Identical variance	AdaptiveManifold homogenizes
Low-friction attractor	XII	Inverted on general n	Mersenne-specific finding
Universal friction ratio	XII	AUC = 0.309	Below random guessing
NRCI scaling law (absolute)	XII-XII I	Decay at -0.00011/bit	Signal converges to 1.0
Tenacity Law / Sovereign Primes	XIII	Contradicted	Primes at higher pressure
Golay normalisation filter	XIII	No improvement	Slope identical to raw
Factor size estimation (general)	XIV	$r = 0.202$	Features encode $\log(n)$, not factors
Spectral centroid blind test	XIV	Regression to mean	Mersenne-specific only

Table 2. Comprehensive list of failed hypotheses across the study

7. Recommended UBP System Changes

7.1 Add a Prime-Layer Harmonic Module

The most consequential architectural change is the addition of a Prime-Layer Harmonic Module, specified in Section 5.3, that operates in parallel with the existing Golay/Leech error-correction layer. This module would use the 4D residue fingerprint $[d(17), d(31), d(113), d(127)]$ in mod-144 space as its fundamental representation, with Jaccard metrics for interval and chord analysis. The module addresses

the three-distance ceiling by operating in a completely different mathematical space (set-theoretic rather than binary coding), achieving 18 unique interval distances and meaningful chord differentiation ($r = 0.82$). It is lightweight (four floating-point values per pitch class), requires no error-correction machinery, and is perfectly transposition-invariant. The existing coding layer would continue to handle stability, error correction, and the NRCI metric, while the prime layer would handle all harmonic analysis tasks. This separation of concerns is clean, well-motivated by the study's findings, and does not require any modification to the existing Golay/Leech pipeline.

7.2 Refine the Modulus Specification

The study established that $144 = 12^2$ is the minimal modulus for the Prime-Layer Module, and that the Mersenne/Fermat invariance holds across all multiples of 12. However, the choice of 144 as the specific modulus has a precise justification: the largest relevant residue (127) plus one octave span (12) gives 139, which is less than 144, avoiding wrap-around artifacts. For larger moduli ($1728 = 12^3$, $20736 = 12^4$, etc.), the 4D fingerprint produces more distinct states but the correlation with consonance does not improve (it remains exactly $r = -0.3770$). This means 144 is not just convenient but optimal: it provides the richest state space without introducing redundancy. The UBP system should formally document 144 as the canonical modulus for the Prime-Layer Module, with the understanding that scaling to higher powers of 12 preserves but does not enrich the harmonic signal.

7.3 Do Not Pursue Factor Size Estimation via Rotation Spectra

The evidence is now definitive: rotation spectral features cannot estimate factor sizes for general composites. The Phase XIII finding ($r = 0.80$ on Mersenne composites) was real but domain-specific to the algebraic structure of the $s^2 - 2$ recurrence modulo $2^p - 1$. On general composites, spectral features encode the magnitude of the number ($\log n$), not its factorization. The blind test on Mersenne composites showed complete regression to the mean, confirming that a model trained on general composites cannot transfer to Mersenne structures and vice versa. The UBP system should not invest further effort in factor size estimation via rotation spectra. The $r = +0.797$ correlation between rotation sign changes and factor count remains a genuine topological signal worth further investigation, but it should be framed as a "factor complexity indicator" rather than a "factor size estimator."

7.4 Use Anchor Distance as the Sole Band-Invariant Primality Indicator

Of all the UBP primality metrics tested across 18 numeric bands, only anchor distance (the Hamming distance from a number's Golay-decoded codeword back to the nearest valid codeword) maintained a stable positive correlation with primality in every band, ranging from $r = +0.312$ to $r = +0.382$ with a mean of $+0.339$. All other metrics (NRCI, Gamma, syndrome weight, rotation sign changes) showed range-dependent inversion, oscillation, or complete collapse. This makes anchor distance the only metric suitable for building range-independent primality heuristics from the UBP coding layer. However, $r = 0.34$ is far too weak for practical primality testing, so this indicator is best understood as a structural feature of the encoding rather than a replacement for deterministic primality tests.

8. Prime Number Method: Reevaluation

8.1 What the UBP Adds to Prime Number Theory

The UBP system contributes three things to the study of prime numbers, none of which is a primality test or factorization algorithm but all of which are structurally interesting. First, the 4D residue fingerprint provides a new geometric representation of integers in mod-144 space that connects Mersenne and Fermat numbers through the XOR bridge at 96. This representation is not useful for distinguishing primes from composites ($r = -0.009$), but it reveals structural properties of special number families that are invisible in standard representations. Second, the rotation sign-change metric during LL iteration provides a topological measure of factorization complexity ($r = 0.80$ with factor count on Mersenne numbers), which could potentially be developed into a theoretical tool for understanding the structure of LL sequences rather than as a practical algorithm. Third, the rank-order invariance of NRCI across bit widths ($r = -0.407$ at every width from 12 to 64 bits) suggests that the UBP encoding preserves ordinal information about primality even as the cardinal signal decays, a property that might have theoretical significance for understanding how coding-theoretic structures interact with multiplicative number theory.

8.2 What the UBP Does NOT Add

The UBP system does not provide a competitive primality test, factorization algorithm, or factor size estimation method. The strongest primality indicator (anchor distance, $r = 0.34$) is far weaker than even trial division. The Lock Pressure landscape does not exhibit the predicted Sovereign Prime structure. The friction ratio is anti-correlated with primality ($AUC = 0.309$). The spectral features of rotation trajectories encode the magnitude of numbers, not their factorization. The coding-layer NRCI signal decays with bit width and is dominated by the magnitude of the number. In short, the UBP system's relationship to prime numbers is descriptive and structural, not algorithmic. It provides a new lens for visualising and understanding certain properties of integer sequences, but it does not solve any computational problem in prime number theory.

8.3 The Constructive Role: 12-TET Decomposition

Perhaps the most intellectually satisfying result of the study is the decomposition of 12-tone equal temperament into Mersenne and Fermat components. The fifth (7 semitones) equals $2^3 - 1$, a Mersenne number, and the fourth (5 semitones) equals $2^{2^0} + 1$, the Fermat prime $F_0 = 3$, minus one octave adjustment. Together, $7 + 5 = 12$: the octave decomposes into a Mersenne component and a Fermat component. This is not a UBP invention but a mathematical fact that the UBP framework made visible by connecting the Circle of Fifths encoding to the mod-144 residue structure. The constructive role of the UBP system here is not to prove new theorems about primes but to provide a unified framework in which the relationships between musical tuning, coding theory, and special number families become geometrically and algebraically apparent. This is the system's genuine contribution: it is a representational framework, not a computational tool.

9. Recommendations for Further Testing

9.1 Scale Testing: 10^4 to 10^6

All primality-related findings were tested in the range [1000, 10000), which is extremely small by number-theoretic standards. The range-dependent inversion documented in Phase XIV shows that UBP metrics change behavior dramatically across sub-ranges within this narrow interval. Whether the anchor distance correlation ($r = 0.34$) persists at 10^5 , 10^6 , or beyond is completely unknown. The Tenacity Law hypothesis might manifest at larger scales where the Lock Pressure landscape has more room to develop local structure. The rank-order NRCI invariance ($r = -0.407$ across 12-64 bits) should be tested at 128, 256, and 512 bits to determine whether the invariance holds or eventually breaks. These scale tests are computationally expensive but essential for establishing whether any of the observed signals are fundamental properties of the encoding or artifacts of the narrow test range.

9.2 The Active-Dimension Signal in the Leech Lattice

Phase IX found that the Leech lattice preserves interval information at $r = 0.8429$ through active dimension sets (which coordinates are non-zero) rather than through Euclidean distances. This signal was not pursued in subsequent phases. The active-dimension representation is essentially a binary pattern of dimension usage, and it might be possible to build a harmonic module based on which Leech dimensions are activated by different pitch classes. This would connect the coding layer (Leech lattice) to the harmonic signal without going through the Golay bottleneck, potentially providing a new pathway for chord differentiation. The computational cost is higher than the 4D fingerprint approach, but the 24D active-dimension pattern might carry information that the 4D mod-144 fingerprint misses.

9.3 Formal Proof of the Invariance Theorem

The empirical invariance of the Mersenne/Fermat residue pattern across moduli (tested at 11 values from 144 to 2,985,984) strongly suggests a formal theorem, but no rigorous proof was constructed during the study. A formal proof would establish: (1) For all primes $p \geq 5$, $2^p - 1 \bmod 12$ is an element of $\{3, 7\}$; (2) For $p \geq 5$ and modulus $m = 12^k$ with $k \geq 2$, $2^p - 1 \bmod m$ is congruent to either 31 or 127 modulo 144 (or their equivalents in the larger modulus); (3) The musical consequence that the fifth and fourth are structurally encoded in the binary expansion of Mersenne and Fermat numbers. The empirical evidence is overwhelming, but a formal proof would place this finding on rigorous mathematical footing and potentially reveal deeper connections to the distribution of primes in arithmetic progressions.

9.4 Cross-Validation with Other Error-Correcting Codes

The study exclusively used the Golay [24,12,8] code and its extensions (Leech, Barnes-Wall). Other error-correcting codes with different distance structures might produce different harmonic properties.

For example, the extended quadratic residue code of length 48, the ternary Golay code [12,6,6], or Reed-Muller codes have different weight enumerators and might provide more than two useful distance buckets for interval classification. Testing whether the three-distance ceiling is specific to the binary Golay code or is a general property of short codes would clarify whether the Prime-Layer Module is necessary because no code can do the job, or because the specific choice of Golay is suboptimal for harmonic analysis. If another code provides three or more useful distance buckets while maintaining error-correction capability, the architectural argument for a separate prime layer would need to be revised.

9.5 Rotation Topology Beyond Lucas-Lehmer

The rotation sign-change signal ($r = 0.80$ with factor count) was only tested with the Lucas-Lehmer recurrence $s = s^2 - 2$. Other recurrences used in primality testing, such as the Fermat primality test ($a^{n-1} \bmod n$), the Miller-Rabin witnesses, or Pollard rho ($x = x^2 + c \bmod n$), might produce different or complementary topological signals when tracked through the 4D residue fingerprint. Each recurrence has different algebraic structure, and the interaction between that structure and the mod-144 geometry might reveal different aspects of factorization. This is the most speculative but also potentially the most rewarding direction for further investigation, as it could establish rotation topology as a general framework for understanding primality tests rather than a curiosity specific to LL.

10. Phase-by-Phase Correlation Atlas

The following table provides a compact reference to every statistically significant correlation discovered across the 14 phases, organised by the type of signal measured and the domain in which it operates. Entries marked "FAILED" indicate the strongest result achieved for that hypothesis, which in all cases was too weak or inconsistent to support the claimed effect. This atlas is intended as a quick-reference for evaluating any future hypothesis against the established baseline of what has already been tested.

Phase	Signal	Best r	Domain	Note
I-II	CoF Gray interval	+0.867	Golay	75.2% R-squared
III	Chord XOR	0.000	Golay	Complete failure
IV	Best permutation	+0.977	Golay	Degenerate (6-point)
V	CoF structural	+0.87	Theory	Gray adjacency chain
VI	Leech Euclidean	+0.13	Leech	Signal destroyed
VII	Full 12-bit seeds	+0.87	Golay	Identical to 4-bit
VIII-D	JI exponents	+0.961	Number theory	Best ever

VIII-H	Spectral chords	-0.619	Spectral	Inverted, strongest chord
IX-E	4D fingerprint chords	-0.879	Prime residue	Euclidean distance
X	Jaccard prox_10 chords	+0.824	Prime residue	First positive > 0.5
XI	NRCI prime vs comp	0.921 vs 0.735	Coding layer	Error-correction signal
XII	Rotation vs factors	+0.797	Topological	Mersenne-specific
XIII	Spectral centroid (M)	+0.796	Frequency	Mersenne composites
XIII	NRCI rank-order	-0.407	Coding layer	Bit-width invariant
XIV	Anchor distance	+0.339	Coding layer	Band-invariant
XIV	Factor estimation	+0.202	Regression	No power

Table 3. Phase-by-phase correlation atlas for all 14 phases